



Saint Columban College
College of Computing Studies
Pagadian City



Fixme2.py

fixme2.py 



Easy

General Skills

Beginner picoMini 2022

Python

AUTHOR: LT 'SYREAL' JONES

Description

Fix the syntax error in the Python script to print the flag.

[Download Python script](#)

Hints 

1 2 3 4

63,813 users solved



85%

Liked



picoCTF{FLAG}

Submit
Flag

Author: The Analyst: Hyposelenia

Challenge: Fixme2.py

Category: General Skills

Date: 01/14/26



I. Objective

Fix the syntax error in the provided Python script so that it runs correctly and prints the flag.

II. Background

Python distinguishes between **assignment** and **comparison** using different symbols. Confusing these symbols can cause logic or syntax errors. This challenge reinforces the importance of understanding basic operators and recognizing when code logic does not need to be deeply analyzed to solve a problem.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **Programiz (Online Python Interpreter)** – Used to quickly test and debug the script

IV. Methodology

1. Downloaded and opened the provided fixme2.py file
2. Ran the script using a Python interpreter
3. Observed that the program did not behave as expected
4. Reviewed the hints provided by the challenge
5. Identified incorrect usage of the equality operator



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```
main.py
3
4
5
6 def str_xor(secret, key):
7     #extend key to secret length
8     new_key = key
9     i = 0
10    while len(new_key) < len(secret):
11        new_key = new_key + key[i]
12        i = (i + 1) % len(key)
13    return ''.join([chr(ord(secret_c) ^ ord(new_key_c)) for (secret_c,new_key_c) in zip
14                    (secret,new_key)])
15
16 flag_enc = chr(0x15) + chr(0x07) + chr(0x08) + chr(0x06) + chr(0x27) + chr(0x21) + chr(0x23)
17             + chr(0x15) + chr(0x58) + chr(0x18) + chr(0x11) + chr(0x41) + chr(0x09) + chr(0x5f) +
18             chr(0x1f) + chr(0x10) + chr(0x3b) + chr(0x1b) + chr(0x55) + chr(0x1a) + chr(0x34) + chr
19             (0x5d) + chr(0x51) + chr(0x40) + chr(0x54) + chr(0x09) + chr(0x05) + chr(0x04) + chr
20             (0x57) + chr(0x1b) + chr(0x11) + chr(0x31) + chr(0x0e) + chr(0x51) + chr(0x5c) + chr
21             (0x44) + chr(0x51) + chr(0x0a) + chr(0x5b) + chr(0x5a) + chr(0x19)
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VI. Explanation

One of the key hint states:

“Are equality and assignment the same symbol?”

This hint points out a very common beginner mistake in Python.

- The **assignment operator (=)** is used to **store a value in a variable**
- The **equality operator (==)** is used to **compare two values**

Example:

```
x = 5    # assignment  
x == 5   # comparison
```

In the script, the code mistakenly used `==` when it should have used `=`. This caused the program to compare values instead of assigning them, preventing the script from working correctly. Once the operator was fixed, the program executed as intended.

Another hint states:

“The `str_xor` function does not need to be reverse engineered for this challenge.”

This means that **you do not need to understand or analyze how the XOR function works internally**. While `str_xor` is responsible for decrypting the flag, it is already correctly implemented. The problem was **not** with the encryption logic, but with a **simple syntax mistake** elsewhere in the code.

For beginners who read this like me:

- *Reverse engineering* would mean analyzing how the function encrypts/decrypts data step-by-step. Do you know that your house was engineered? Reverse engineering in that instance is when you analyze what your house is made of and how your house is made. Another explanation, we destroy something just to understand how it works.



- In this challenge, doing so is unnecessary because fixing the syntax error alone allows the function to run and reveal the flag automatically

In short: **fix the syntax first, don't overthink the crypto.**

VII. Conclusion

Fixme2.py teaches an important programming lesson: small syntax differences can completely break a program. Understanding the difference between assignment (=) and comparison (==) is essential in Python.

Additionally, the challenge emphasizes that not every problem requires deep code analysis—sometimes the simplest fix is all that's needed.

— **The Analyst: Hyposelenia**